Bayesian Modelling of River Water Quality

Censoring, Flow and Source Contributions

Fakhruddin Hakim Hussain

MSc Dissertation, University of Edinburgh

October 2026

1 Abstract

This MSc dissertation studies how routine river-flow and water-quality observations can inform a model-based decomposition of point-source and diffuse-source concentration. The analysis reports 16,233 valid concentration readings at six English river sites, covering ammonia, nitrate, orthophosphate and dissolved zinc. Daily aggregation retains exact, left-censored and mixed interval-censored information, rather than replacing below-limit observations by arbitrary concentrations.

A classical Load Apportionment Model (LAM) supplies a constrained concentration–flow relationship. Generalised additive models (GAMs) provide flexible benchmarks. The final Bayesian models partially pool site-specific source levels within each pollutant, share flow exponents and include an annual sine–cosine effect. A time-varying extension adds linear trends to the logarithms of the source coefficients. Later observations in each site–pollutant series form a chronological 20% test block.

The dissertation reports site-averaged log-concentration RMSE reductions of 30.5% for zinc and 19.9% for orthophosphate relative to static Bayesian models. The dynamic model worsens ammonia and brings little benefit for nitrate. The wider baseline comparison is important: zinc's dynamic RMSE of 0.164 is below the best GAM's 0.180, while orthophosphate's dynamic RMSE of 0.282 remains above the flow-based GAM's 0.200.

Posterior source shares represent the two assumed flow components, not directly measured physical discharges. A reported prior-sensitivity study finds more variation in site-level trend estimates than in most concentration predictions, with weaker mixing under tighter priors.

This report audits and presents the dissertation findings. Sixty main-table numeric entries were checked against the source, together with the archived code and reported sensitivity table. The available R scripts are earlier per-site revisions. The research decision is to restore the frozen split and final hierarchical implementation before extending the model. Missing data, final code and fit artifacts prevent a new run; no regenerated performance or convergence claim is made.

2 Introduction

Concentration depends on both pollutant input and the volume of water carrying it. A relatively steady discharge can produce dilution as flow increases, while rainfall can mobilise material across a catchment. These patterns motivate a statistical decomposition into a decreasing and an increasing concentration component.

The Load Apportionment Model uses two flow-dependent power functions to represent these behaviours [2]. It needs paired concentration and flow observations, making it attractive for routine monitoring records. Its simplicity also imposes strong assumptions: observed behaviour need not map uniquely to particular treatment works, farms or other physical sources.

The dissertation addresses three statistical difficulties. Detection limits make some concentrations only partially observed. Irregular sampling can give heavily sampled days excessive weight. Long monitoring records can contain changes in environmental conditions, source behaviour and laboratory reporting, while sparse records can leave separate source estimates unstable.

The modelling sequence is therefore deliberate. First construct daily observations with defensible censoring bounds. Then establish classical and flexible baselines. Finally, use hierarchical Bayesian models to share information across sites and allow interpretable temporal change. The purpose is to assess both concentration prediction and the uncertainty of the source decomposition.

The central question is when the extra structure helps. A dynamic model should earn its complexity through predictive behaviour and diagnostics. Improvement relative to one Bayesian baseline does not imply superiority to every classical or additive alternative. The full reported comparison is retained here, including pollutants for which the more complex specification is unsuccessful.

3 Monitoring Data and Study Design

3.1 Sites, pollutants and reported counts

Concentration observations came from the Environment Agency Water Quality Archive, while daily mean flow came from its Hydrology Data Explorer [5]. Six locations are reported: Thames at Teddington, Severn at Haw Bridge, Tees at Low Worsall, Eden at Sheepmount, Trent at Dunham and Wensum at Sweet Briar Road Bridge.

The four pollutants are ammonia, nitrate, orthophosphate and dissolved zinc. Eden has only one zinc observation, itself censored, and that combination is excluded. This leaves 23 usable site–pollutant series from 24 possible combinations.

The dissertation reports 16,233 valid raw concentration records and 13,227 daily records: 10,816 exact, 2,280 all-censored and 131 mixed. There are 914 days with multiple samples, containing 3,920 raw records. The arithmetic $16,233 - (3,920 - 914) = 13,227$ reconciles daily aggregation. The exact number retained after flow matching and group exclusions requires the missing frozen dataset; these counts are not presented as a verified final fitted-row total.

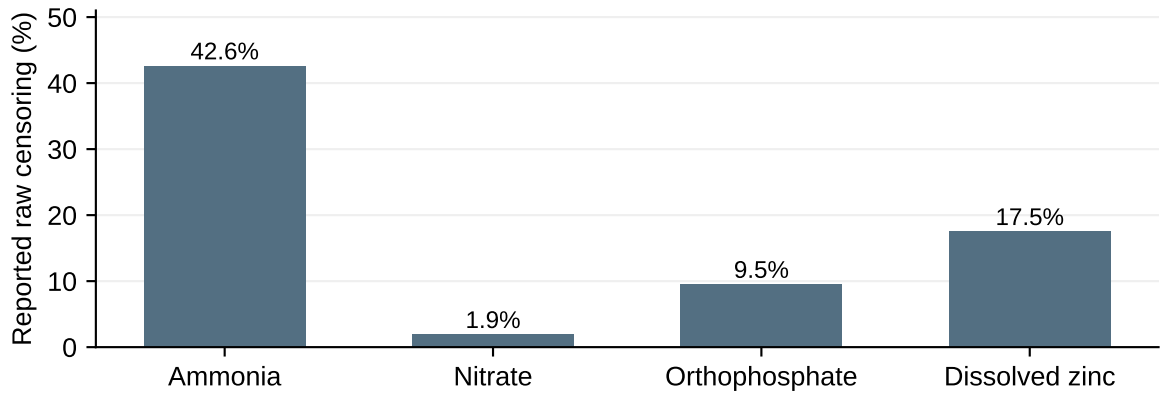


Figure 1: Reported censoring rates among raw readings, by pollutant. These percentages are not the daily censoring rates or a pooled rate across all pollutants. Ammonia is especially affected.

3.2 Matching and temporal evaluation

Daily concentration records are matched to positive daily mean flow by site and date. This reduces unequal within-day sampling weight, but daily mean flow may differ from flow at the actual sampling time. Monitoring frequency and detection limits also vary across locations and years.

The reported split orders each series by date, using its earliest 80% for training and latest 20% for testing. The same observations support comparisons across models. This is conditional prediction of concentration given contemporaneous flow, rather than an operational forecast of unknown future flow and concentration. A common historical cutoff across all pooled sites is not documented in the available final artifacts.

4 Daily Aggregation and Censored Likelihood

4.1 Keeping the information in detection limits

For a day with n samples, let S_E be the sum of exact values and S_U the sum of upper limits for censored values. Daily means are constructed before taking logarithms:

$$\begin{aligned} &C_{\text{day}} = S_E/n, && \text{all exact,} \\ &0 < C_{\text{day}} \le S_U/n, && \text{all censored,} \\ &S_E/n < C_{\text{day}} \le (S_E + S_U)/n, && \text{mixed.} \end{aligned} \tag{1}$$

For example, one exact value of two and one measurement below one imply a daily mean between one and 1.5. Replacing the second value by 0.5 would collapse that interval to a particular number without evidence. The reported workflow carries the interval into the likelihood instead.

4.2 Observation model

Let $Y_i^* = \log_{10} C_i$ and assume $Y_i^* \sim \mathcal{N}(\mu_i, \sigma_i^2)$. On this log scale, an exact value is y_i, a left-censoring bound is u_i, and an interval has bounds $(l_i, u_i]$. The likelihood contribution is

$$L_i = \begin{cases} \sigma_i^{-1}\phi((y_i - \mu_i)/\sigma_i), & \text{exact,} \\ \Phi((u_i - \mu_i)/\sigma_i), & \text{left-censored,} \\ \Phi((u_i - \mu_i)/\sigma_i) - \Phi((l_i - \mu_i)/\sigma_i), & \text{interval-censored.} \end{cases} \tag{2}$$

Exact observations contribute a density, while censored observations contribute a probability mass over the known range. Censoring therefore changes the observation model; it is not merely a missing-value imputation step.

The distribution is Gaussian on log concentration. Consequently, exponentiating μ_i gives a conditional median under this observation model, rather than the arithmetic mean concentration. A concentration-scale expectation would need the corresponding lognormal variance adjustment. The source-apportionment interpretation should distinguish the latent concentration curve from that expectation.

All fitting and evaluation must use the same concentration units and logarithm convention within a pollutant. A continuous density can exceed one, so a negative predictive NLL is possible. It is not a negative probability.

5 Classical and Additive Benchmarks

5.1 The two-component concentration curve

For positive flow Q, the classical model is

$$C(Q) = AQ^{B-1} + MQ^{D-1}, \qquad A, M > 0, \quad 0 < B < 1 < D. \tag{3}$$

The first component decreases with flow and the second increases. The classical mean log concentration is $\log_{10} C(Q)$. Unconstrained numerical parameters are transformed with exponentials and a logistic function to enforce the stated restrictions.

The archived classical script maximises the censoring-aware likelihood using several fixed starting values. The dissertation reports optimisation convergence code zero for all 23 fits, while also noting boundary estimates and weakly supported components. A numerical stopping flag does not establish that both components are identifiable.

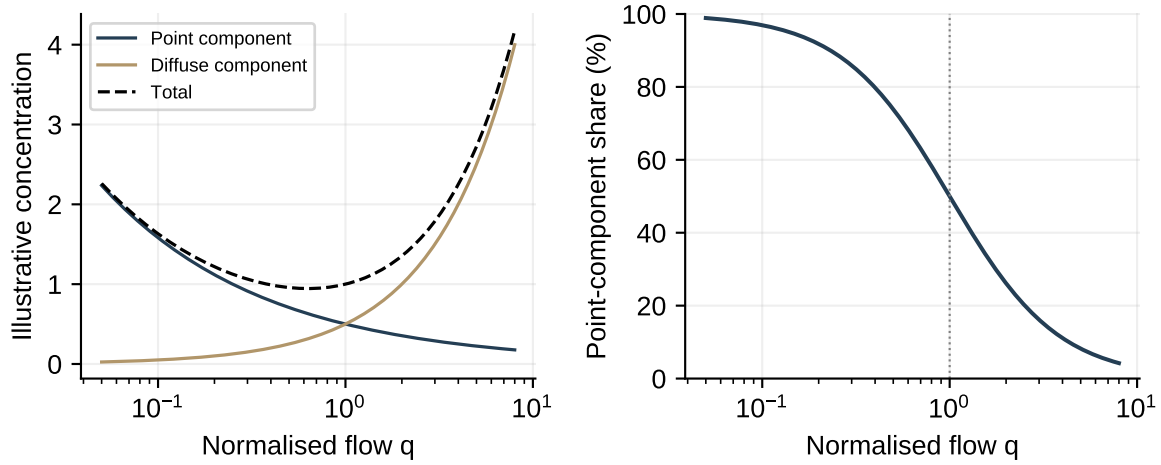


Figure 2: Illustrative model geometry, not a fitted river result. The chosen values are $A = M = 0.5$, $B = 0.5$ and $D = 2$. The left panel shows the two components and total; the right shows the resulting point-component share.

5.2 Flexible prediction controls

A time–season GAM uses a smooth long-term time effect and cyclic annual seasonality. A second GAM adds a smooth effect of log flow:

$$\mu_i = \beta_0 + f_t(t_i) + f_d(d_i) \quad \text{or} \quad \mu_i = \beta_0 + f_t(t_i) + f_d(d_i) + f_Q(\log_{10} Q_i). \tag{4}$$

The reported fits use censored-Normal likelihoods in `mgcv`, with REML smoothness selection. They predict concentration without imposing a physical point/diffuse allocation. Their role is to test whether source interpretability sacrifices predictive accuracy and whether flow adds useful information beyond time and season.

6 Hierarchical Bayesian Source Models

6.1 Static seasonal specification

The final dissertation fits sites jointly within each pollutant. With normalised flow $q_{is} = Q_{is}/Q_{\text{ref}}$, site-specific source coefficients satisfy

$$A_s = \exp(\alpha_A + u_{A,s}), \qquad M_s = \exp(\alpha_M + u_{M,s}). \tag{5}$$

The site deviations have hierarchical distributions with estimated between-site scales. Partial pooling retains differences between sites while sharing information within a pollutant. It does not force every site to have the same source levels.

Flow exponents are shared within each pollutant, using $B = \text{logit}^{-1}(b)$ and $D = 1 + \exp(d)$. The static seasonal mean is

$$\begin{aligned} \mu_{is} &= \log_{10}\{A_s q_{is}^{B-1} + M_s q_{is}^{D-1}\} + h(d_{is}), \\ h(d) &= \gamma_{\sin} \sin(2\pi d/365.25) + \gamma_{\cos} \cos(2\pi d/365.25). \end{aligned} \tag{6}$$

The shared sine–cosine terms represent a repeating annual effect. Site-level source sizes can differ, while the flow-response exponents and seasonal coefficients remain common within the pollutant.

6.2 Linear time variation in source coefficients

For centred and scaled time z, the extension is

$$\begin{aligned} \log A_s(z) &= \alpha_A + u_{A0,s} + (\beta_A + u_{A1,s})z, \\ \log M_s(z) &= \alpha_M + u_{M0,s} + (\beta_M + u_{M1,s})z. \end{aligned} \tag{7}$$

The source coefficients remain positive. Linear trends on their natural-log scales imply multiplicative change over time; this is not a random-walk state process or a flexible temporal spline in the final model. The exponents B, D remain fixed through time.

The final analysis is described as fitted in R with `brms`/Stan [3]. The available Bayesian scripts are earlier per-site models with different parameterisations and seasonal/time smooths. Their chain settings and results must not be substituted for the final hierarchical fit. The final code, residual-scale predictor and complete prior parameterisation need restoration for an exact rerun.

7 Evaluation and Diagnostics

7.1 Predictive scores and averaging

The dissertation's principal score is test negative log-likelihood, using exact densities and censoring masses as appropriate. For a Bayesian predictive distribution, the likelihood is integrated over posterior draws. A stable Monte Carlo score is

$$\text{NLL} = -\frac{1}{n}\sum_{i} \log\left\{\frac{1}{R}\sum_{r=1}^{R} L_i(\theta^{(r)})\right\}. \tag{8}$$

This is the log of an average predictive likelihood, rather than the average log likelihood. The archived scripts implement a log-mean-exp calculation; the final score cannot be regenerated without its posterior artifacts.

MAE and RMSE use only exact test observations on the $\log_{10}$ concentration scale. Censored observations still inform fitting and likelihood scoring. A heavily censored pollutant can therefore have a point-error assessment that describes only its observed exact-value subset.

The main reported tables average site-level metrics within each pollutant. A mean of site RMSEs is not the same as pooling every squared residual before taking one square root. The headline reductions below are recomputed from the reported rounded site averages. They are error reductions, not reductions in pollution or direct estimates of intervention benefit.

7.2 Residual and posterior checks

For fitted predictive CDF F_i, randomised quantile residuals transform the admissible probability range to a standard-Normal scale:

$$U_i = \begin{cases} F_i(y_i), & \text{exact}, \\ \text{Uniform}(0, F_i(u_i)), & \text{left-censored}, \\ \text{Uniform}(F_i(l_i), F_i(u_i)), & \text{interval-censored}, \end{cases} \qquad r_i = \Phi^{-1}(U_i). \tag{9}$$

Their mean, spread and temporal dependence are useful diagnostic targets [4]. The source reports better average residual calibration for the classical LAM than the GAMs, while some series retain serial dependence. This does not establish ideal calibration for every model or series.

The Bayesian diagnostic framework includes R-hat, bulk and tail effective sample sizes, divergent transitions, tree depth and posterior predictive checks. The report retains the source's quantitative prior-sensitivity diagnostics in Section 10. It does not infer universal convergence from the presence of Bayesian code or credible intervals.

8 Reported Predictive Results

8.1 Static versus time-varying Bayesian models

Table 1: Dissertation Table 5, printed page 12. These are reported findings, not newly fitted results. Lower scores are better; point errors are in log10 concentration.

Pollutant	Bayesian model	NLL	MAE	RMSE
Ammonia	Static	1.305	0.920	1.031
	Time-varying	1.362	1.288	1.422
Nitrate	Static	-0.836	0.077	0.101
	Time-varying	-0.845	0.083	0.103
Orthophosphate	Static	0.693	0.314	0.352
	Time-varying	0.293	0.244	0.282
Dissolved zinc	Static	0.200	0.208	0.236
	Time-varying	-0.400	0.126	0.164

For zinc, RMSE falls from 0.236 to 0.164: a 30.5% reduction from rounded entries, conventionally described as approximately 31%. Orthophosphate falls from 0.352 to 0.282, a 19.9% reduction. All three scores improve in these two comparisons.

Ammonia moves in the other direction. Its RMSE increases by 37.9%, and both MAE and NLL worsen. The source discusses heavy censoring and abrupt changes, but the available evidence does not isolate a single failure mechanism.

Nitrate has a small NLL improvement of 0.009 alongside slightly worse MAE and RMSE. The choice of score therefore matters. The results favour a simpler static description for nitrate, rather than a claim that temporal complexity helps every pollutant.

These are comparisons of the reported Bayesian specifications. The prior table also changes some scale priors and lists a different residual-scale parameterisation, so the documentation does not establish a one-change-only time-trend ablation.

8.2 Comparison with all classical and additive controls

Table 2: Mean site RMSE from dissertation Tables 4 and 5. GAM-T includes time/season; GAM-Q also includes flow. B-Static and B-Time are the Bayesian models.

Pollutant	LAM	GAM-T	GAM-Q	B-Static	B-Time
Ammonia	0.298	0.335	0.371	1.031	1.422
Nitrate	0.104	0.267	0.150	0.101	0.103
Orthophosphate	0.252	0.235	0.200	0.352	0.282
Dissolved zinc	0.192	0.205	0.180	0.236	0.164

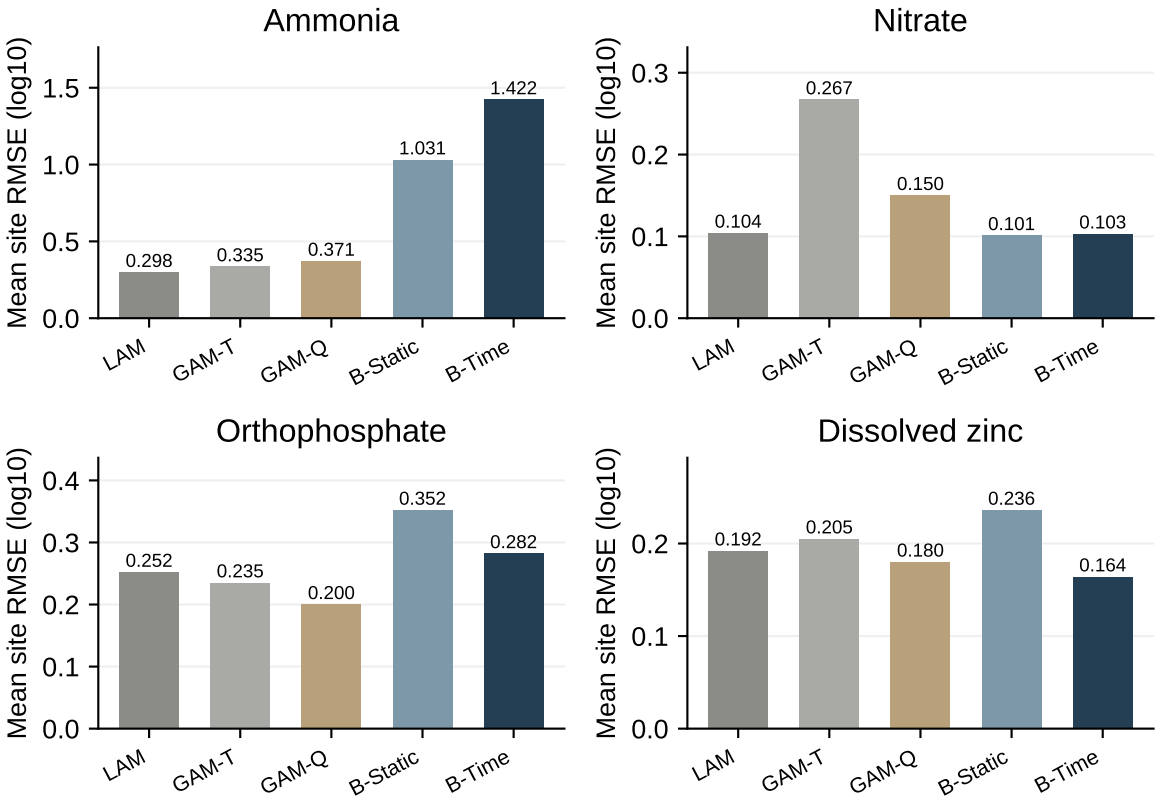


Figure 3: The wider comparison changes the interpretation of a gain over static Bayes. Vertical scales differ by pollutant. Values are the reported rounded site averages; no uncertainty bars or significance claims are inferred.

Zinc's dynamic model has the lowest reported RMSE of these controls, about 8.9% below the best non-Bayesian GAM. Orthophosphate's dynamic RMSE is 41.0% higher than the flow-based GAM's 0.200, despite its improvement over static Bayes. Classical LAM is the strongest of these point-error models for ammonia, while static Bayes is marginally best for nitrate.

Orthophosphate's dynamic NLL of 0.293 is nevertheless slightly lower than the best non-Bayesian NLL of 0.305. Density scores and point errors can rank models differently.

Adding flow improves GAM NLL in 18 of 23 series: six ammonia, five nitrate, five orthophosphate and two zinc. Counts and mean scores can disagree: orthophosphate's mean GAM NLL rises from 0.305 to 0.405 despite five of six series improving.

9 Source Interpretation and Identifiability

For a site, time and flow, the point-component share is

$$s_P(q,z) = \frac{A_s(z)q^{B-1}}{A_s(z)q^{B-1} + M_s(z)q^{D-1}}, \qquad s_D = 1 - s_P. \tag{10}$$

At reference flow $q = 1$, it reduces to $A_s(z)/(A_s(z) + M_s(z))$. A common seasonal multiplier cancels from the share. The log odds satisfy

$$\log \frac{s_P}{1 - s_P} = \log A_s(z) - \log M_s(z) + (B - D)\log q. \tag{11}$$

Thus the relative share can change because one component changes faster than the other, even if both move in the same direction. At fixed time, the assumed $B < D$ forces the point share to fall with flow. That shape is part of the model, not separate physical confirmation of the source labels.

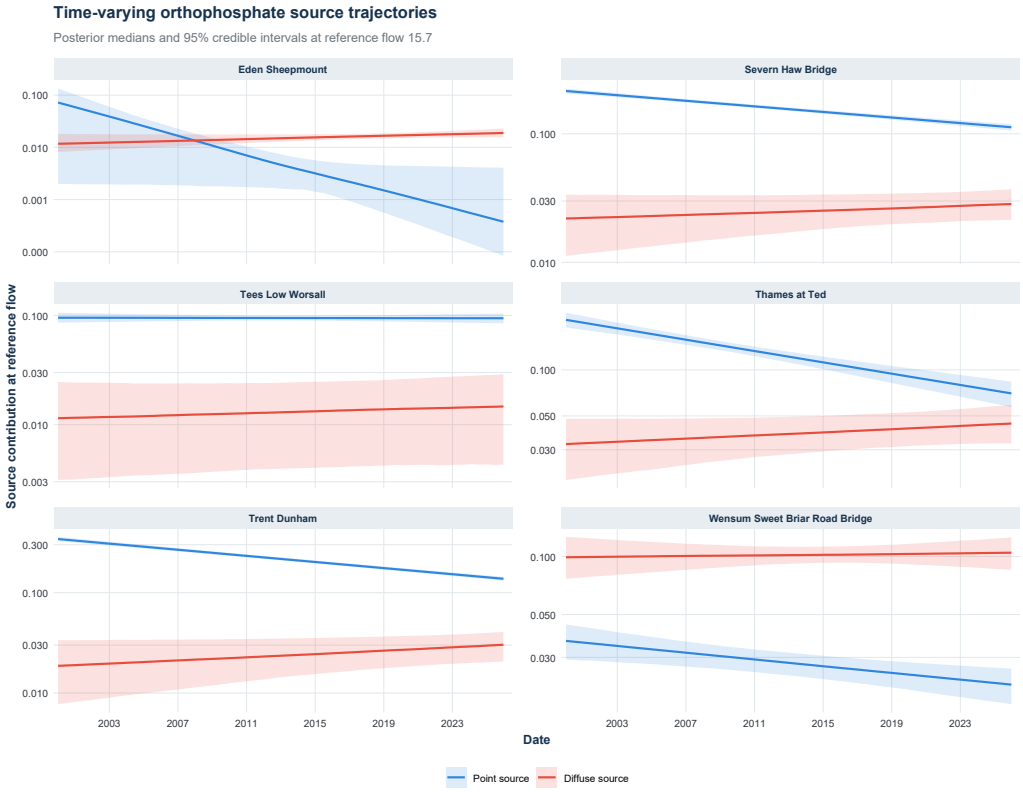


Figure 4: Reported orthophosphate component trajectories, reproduced from dissertation Figure 6. These are the original model summaries, not regenerated posterior draws or test forecasts. Blue denotes the point component and red the diffuse component; ribbons are the reported 95% credible intervals. Reference flow is 15.7 in the original flow units.

Reported orthophosphate paths suggest declining point contributions at several sites. Only 12 of 23 classical crossover flows lie inside the observed training-flow range, and several fits approach parameter bounds. Good total-concentration prediction can coexist with uncertain source allocation.

Source-share credible intervals describe uncertainty under this model. They are not direct physical attribution or new-sample prediction intervals. The Wensum zinc reporting shift around 2015 also shows how laboratory change could be mistaken for environmental change.

10 Prior Sensitivity and Sampling Evidence

The dissertation's focused sensitivity analysis concerns the time-varying orthophosphate model. Prior means are held fixed, with prior standard deviations multiplied by 0.67 or 1.5. The following values are reported in its Table 3, rather than recalculated from new posterior draws.

Table 3: Reported sensitivity and sampling diagnostics. Prediction changes are on the log10 concentration scale, relative to the baseline prior.

Measure	Baseline	Tighter	Wider
Maximum R-hat	1.010	1.025	1.004
Minimum bulk ESS	454	137	725
Minimum tail ESS	458	763	1055
Divergent transitions	0	0	0
Median B	0.487	0.461	0.493
Median D	1.927	1.842	1.956
Median between-site slope SD	1.273	0.489	1.562
Mean absolute prediction change	–	0.043	0.010
Maximum absolute prediction change	–	1.152	0.182
Prediction correlation	–	0.9870	0.9996

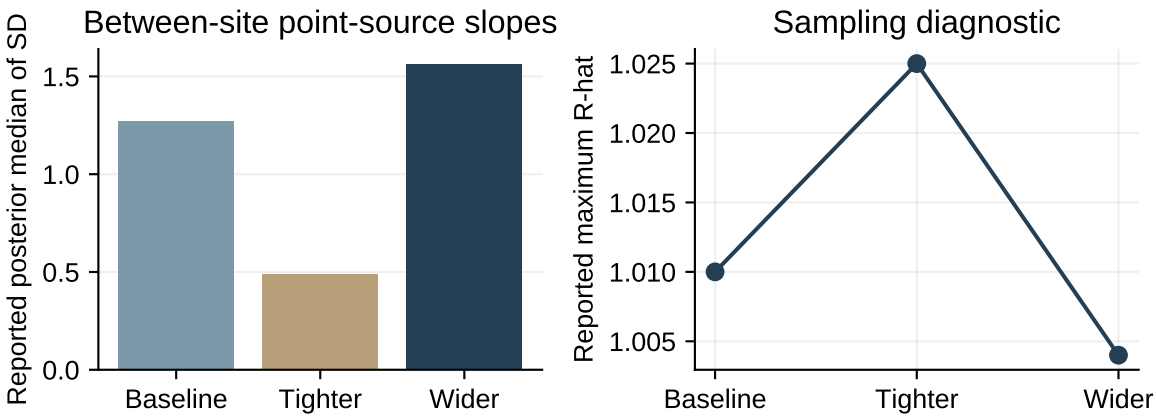


Figure 5: Reported sensitivity of between-site point-source trend variation and the maximum R-hat. Stable concentration predictions and stable source-trend estimates are different requirements.

Wider-prior predictions correlate 0.9996 with baseline predictions, with mean absolute change 0.010. Their maximum change is still 0.182. Tighter priors change 12.5% of predictions by more than 0.1, compared with 1.0% under wider priors. Correlation alone therefore does not establish uniformly negligible sensitivity.

The between-site slope SD changes from 1.273 to 0.489 under tighter priors and to 1.562 under wider priors. The tighter fit also has weaker mixing, with maximum R-hat 1.025 and minimum bulk ESS 137. All three reported divergence counts are zero, but that one statistic does not establish convergence. These are source-reported diagnostics; final draws are absent for an independent audit.

11 Reproducibility, Limitations and Next Milestone

11.1 Available evidence and checks completed

The final dissertation and its main tables were inspected. Sixty numeric entries across Tables 4 and 5 were extracted and checked, including all 24 Bayesian entries against the repository transcription. The sensitivity table was checked separately. The Library code archive matches all five R scripts in the pinned repository byte for byte. Additional numerical checks concern the stated likelihood and source-share algebra, rather than a new river-data fit.

The revised preflight defaults to final-model scope and exits with a missing-input status. It requires the frozen `model_data_split.rds` and a restoration manifest hashing both that split and the final model code. An explicit archived scope checks only the split file. Three regression tests verify that a split alone cannot pass final-model preflight and that changed or external manifest paths are rejected. This is an input-availability check, not a fit or convergence verification. The final hierarchical code and fit artifacts are also missing. The archived scripts fit earlier per-site models, including temporal/seasonal splines that differ from the final linear-trend, sine–cosine hierarchy. Running them on newly downloaded data would not reproduce Table 5.

11.2 Unresolved specification and validation details

The final prior table describes a Normal distribution's second argument as a standard deviation, while entries use squared-scale notation. It also lists time-varying residual-scale coefficients without specifying their full predictor and link. Those ambiguities need the final code; this report does not silently choose an interpretation and call it the original fitted model.

Exact post-aggregation and post-flow-match row counts, observation-date ranges, training-only reference scales and per-site split dates require the frozen dataset. Because the hierarchy pools sites, per-series chronological ordering alone does not establish that every contributing training observation precedes every test prediction in calendar time. A common historical cutoff must be checked before claiming a real-time multi-site forecasting experiment. No observed cross-site leakage is asserted without those dates.

Serial dependence, changing censoring limits and reporting methods also limit source interpretation. The existing test results have already been examined, and they do not provide an untouched model-selection-and-confirmation design. Neither the numerical uncertainty in the reported RMSE differences nor a causal environmental effect can be reconstructed from rounded tables alone.

11.3 A concrete next milestone

Restore the frozen split, final model specification and posterior/predictive artifacts. Resolve prior and residual-scale notation before fitting. Reproduce one complete pollutant comparison first, retaining classical LAM and both GAM controls, with one documented calendar information boundary across sites. Export observation-level predictive scores, exact-case errors, source shares and all sampler diagnostics. The deliverable should be a versioned input manifest, executable R workflow, prediction ledger and a table that reconciles to the dissertation or explains each difference. This restoration is proposed work.

12 Conclusion

The dissertation combines censoring-aware data handling, interpretable concentration–flow models and hierarchical Bayesian source estimation. Its important modelling choices include retaining detection-limit information, limiting unequal within-day sampling weight, partially pooling site effects and distinguishing source-size change from fixed flow-response exponents.

The reported dynamic Bayesian comparison helps zinc and orthophosphate relative to static Bayes, while ammonia worsens and nitrate gains little. The full control comparison is more selective: dynamic zinc has the lowest reported point error, but the flow GAM is better for orthophosphate. Model complexity and scientific interpretability must be assessed against those stronger controls.

The work demonstrates statistical formulation and critical model comparison. Its source shares remain conditional model components. This portfolio report preserves the reported positive and adverse findings, the uncertainty analysis and the precise evidence required for a reproducible next stage.

13 Sources and Report Provenance

The primary source is the MSc dissertation Censoring-Aware and Time-Varying Load Apportionment Modelling of River Water Quality by Fakhruddin Hussain, University of Edinburgh. The original declaration is dated 26 June 2026. The source acknowledges supervision by Dr Jordan Richards and Dr Amanda Lenzi, with data and advice from Mike Dunbar. This report's cover date is its assembly date, not a new dissertation-completion date.

The audit uses repository commit `2b8b7b0137bd` [1]. The companion verification JSON records source hashes, table values, arithmetic, code-archive comparisons and missing-artifact status. It explicitly records that final models and posterior diagnostics were not rerun. The LinkedIn entry supplies the project description, relevant skills, public source link and PDF media details.

October 2026 review: the public research decision note records the leading comparison, evidence limits and next experiment. The pinned source versions below identify the experiments; the review changes presentation and the explicitly described code checks.

References

[1] Hussain, F. H. Bayesian river water quality. Versioned dissertation, archived scripts and reproduction status. Main results: dissertation Tables 4 and 5, printed pages 10 and 12.

[2] Bowes, M. J., Smith, J. T., Jarvie, H. P. and Neal, C. (2008). Modelling of phosphorus inputs to rivers from diffuse and point sources. Science of the Total Environment, 395, 125–138. DOI. Methodological reference cited in the dissertation.

[3] Bürkner, P.-C. (2017). brms: An R Package for Bayesian Multilevel Models Using Stan. Journal of Statistical Software, 80(1), 1–28. Paper.

[4] Dunn, P. K. and Smyth, G. K. (1996). Randomized Quantile Residuals. Journal of Computational and Graphical Statistics, 5(3), 236–244. DOI. Diagnostic reference cited in the dissertation.

[5] Environment Agency. Water Quality Archive / Explorer; Hydrology Data Explorer. The dissertation records access in June 2026; no new historical sample was substituted for its frozen input.